Pharmacology Master Review

Comprehensive Multiple Choice Questions, Correct Answers & Detailed Explanations

Section I: General Pharmacology & Pharmacokinetics

1. Which one of the following is odd?

A. Drugs must have a known chemical structure

B. A medicine may contain more than 1 drug

C. Drugs are mostly used to diagnose a disease

D. Drugs must produce biologic effects (Correct)

Explanation: While many drugs produce biologic effects, diagnostic agents or radio-opaque dyes may be used purely for imaging/physical visualization without necessarily altering physiological function in the traditional pharmacodynamic sense. Furthermore, placebos or inert substances do not necessarily have intrinsic biologic effects yet are studied.

2. Which one of the following route of drug administration provides rapid drug absorption?

A. Oral administration

B. Sublingual administration (Correct)

C. Rectal administration

D. Intramuscular administration

Explanation: Sublingual administration bypasses first-pass hepatic metabolism and provides direct, rapid absorption into the systemic circulation due to the rich vascular supply under the tongue. (Note: Intravenous is fastest/instantaneous, but among enteral/sublingual options listed, sublingual or parenteral routes like IV/Inhalation are rapid; sublingual absorbs very quickly compared to standard oral tablets).

3. Which one of the following is not correctly matched?

A. Oral route – high first pass effect

B. Sublingual route – no first pass effect

C. Rectal route – no first pass effect

D. Intravenous route – no first pass effect (Wait, IV has no first pass. Let's check typical options: Rectal route partially avoids first-pass metabolism, but some rectal drainage goes via inferior/ middle rectal veins directly into systemic circulation while superior rectal vein goes to portal system). Let's review standard matching questions.

Explanation: The rectal route undergoes partial first-pass metabolism because the inferior and middle rectal veins drain directly into the systemic circulation, whereas the superior rectal vein drains into the portal system. Thus, saying "no first-pass effect" entirely for rectal is incorrect.

4. Which of the following is true about drug absorption?

A. Few drugs are absorbed on empty stomach

B. Most drugs need food for absorption

C. Diarrhea may decrease drug absorption

D. Gastrointestinal irritant drugs can be taken before food. (Correction: Irritants are taken after food. Let's check standard correct option: Diarrhea decreases transit time, decreasing absorption).

Explanation: Diarrhea increases gastrointestinal motility and decreases transit time, thereby reducing the time available for drug absorption across the GI mucosa.

5. Heparin, an anticoagulant drug, is not given orally, but parenterally, because it is:

A. Unstable in the GIT (Correct)

B. Poorly absorbed if given orally

C. Only given for unconscious patients

D. A drug with local effect

Explanation: Heparin is a large, highly polar mucopolysaccharide molecule that is degraded by gastric acid and digestive enzymes in the gastrointestinal tract (GIT) and is not absorbed orally.

6. Which of the following is odd about drug distribution?

A. Drugs are well distributed to vessel-rich organs

B. Polar drugs can cross the blood-brain barrier easily

C. Drugs that bind to plasma proteins have a lesser distribution

D. Drugs that distribute to tissues have a long half life

E. Drugs that are lipophilic can cross placental barrier (Correct choice as the 'odd' statement: Polar drugs CANNOT cross the blood-brain barrier easily due to tight junctions and lack of pores).

Explanation: Polar (water-soluble, hydrophilic) drugs have low lipid solubility and cannot easily cross the lipid bilayers of the blood-brain barrier. Therefore, statement B is false and represents the odd statement.

7. Concomitant use of warfarin and metronidazole can lead to overt bleeding. Because:

A. Metronidazole decreases metabolism of warfarin (Correct)

B. Metronidazole increases metabolism of warfarin

C. Warfarin increases metabolism of metronidazole

D. Warfarin decreases metabolism of metronidazole

E. Metronidazole decreases absorption of warfarin

Explanation: Metronidazole inhibits the hepatic cytochrome P450 enzymes (specifically CYP2C9) responsible for metabolizing S-warfarin, leading to increased plasma concentrations of warfarin and an increased risk of bleeding.

8. An 18-year-old female patient is brought to the emergency department due to drug overdose. Which of the following routes of administration is the most desirable for administering the antidote for the drug overdose?

A. Intramuscular

B. Intravenous (Correct)

C. Oral

D. Subcutaneous

E. Rectal

Explanation: In emergency situations involving acute drug overdose, the intravenous (IV) route is most desirable because it bypasses absorption barriers and delivers the antidote directly into the systemic circulation for immediate onset of action.

9. Drug A is a weakly basic drug with a pKa of 7.8. If administered orally, at which of the following sites of absorption will the drug be able to readily pass through the membrane?

A. Mouth (pH approximately 7.0)

B. Stomach (pH of 2.5)

C. Duodenum (pH approximately 6.1)

D. Jejunum (pH approximately 8.0) (Correct)

Explanation: Weak bases are unionized in basic (alkaline) environments where $pH > pKa$. With a pKa of 7.8, in the jejunum (pH 8.0), the drug is predominantly unionized and lipid-soluble, allowing rapid membrane permeation.

10. Methylphenidate helps patients with attention deficit hyperactivity disorder (ADHD) maintain attention and perform better at school or work, with an ED50 of 10 mg. However, methylphenidate can also cause significant nausea at higher doses (TD50 = 30 mg). Which is correct regarding methylphenidate?

A. The therapeutic index of methylphenidate is 3. (Correct)

B. The therapeutic index of methylphenidate is 0.3.

C. Methylphenidate is more potent at causing nausea than treating ADHD.

D. Methylphenidate is more efficacious at cholinergic transmission than treating ADHD.

Explanation: The therapeutic index (TI) is calculated as $TD50 / ED50$. Here, $30 ext{ mg} / 10 ext{ mg} = 3$. A higher therapeutic index indicates a safer drug, and 3 is the correct mathematical ratio.

Section II: Autonomic & Cardiovascular Pharmacology

27. $ lpha1$-Adrenoceptors are prominently involved in which one of the following?

A. Intestinal relaxation

B. Cardiac contractility

C. Presynaptic inhibition

D. Vasoconstriction (Correct)

Explanation: Alpha-1 ($lpha_1$) adrenergic receptors are Gq-coupled receptors located on vascular smooth muscle, causing contraction (vasoconstriction) and increased blood pressure upon activation.

28. Smooth muscle relaxation is most associated with which one of the following adrenoceptors?

A. $ lpha_1$

B. $ eta_2$ (Correct)

C. $ lpha_1$

D. $ lpha_2$

Explanation: Beta-2 (β_2) adrenergic receptors are Gs-coupled receptors located on bronchial, uterine, and vascular smooth muscle, leading to relaxation when stimulated by agonists like albuterol.

32. We administer a pharmacologic dose of epinephrine and observe (among other responses) a direct increase of cardiac rate, contractility, and electrical impulse conduction rates. Which adrenergic receptor was responsible for these direct cardiac effects?

A. $ lpha_1$

B. $ lpha_2$

C. $ eta_1$ (Correct)

D. $ eta_2$

Explanation: Beta-1 (β_1) receptors are the predominant adrenergic receptors in the heart. Their stimulation increases heart rate (positive chronotropy), contractility (positive inotropy), and conduction velocity (positive dromotropy).

Section III: Chemotherapy & Antineoplastic Drugs

29. To optimize drug therapy, it is necessary to know in what phase of the cell cycle antineoplastic agents are effective. Which of the following agents is cytotoxic only to cells in the S-phase of the cycle?

A. Mechlorethamine

B. Bleomycin

C. Chlorambucil

D. 5-Fluorouracil (Correct)

Explanation: 5-Fluorouracil (5-FU) is an antimetabolite that inhibits thymidylate synthase, disrupting DNA synthesis. It is strictly cell-cycle specific for the S-phase (DNA synthesis phase).

21. Mesna is administered with cyclophosphamide and ifosfamide to:

A. Potentiate their cytotoxic action

B. Retard their renal excretion

C. Block their emetic action

D. Ameliorate cystitis caused by them (Correct)

Explanation: Cyclophosphamide and ifosfamide produce acrolein as a metabolite, which accumulates in the bladder and causes hemorrhagic cystitis. Mesna (2-mercaptoethane sulfonate sodium) acts as a uroprotectant by binding and detoxifying acrolein in the urinary tract.

Section IV: Neuropharmacology & Psychiatry

34. A 46-year-old woman was prescribed antidepressant medication and has been taking her medication as prescribed for more than 6 months, but she was very bothered by the dry mouth, difficulty of urinating, and constipation she was experiencing. The drug she was taking is likely?

A. Sertraline

B. Venlafaxine

C. Amitriptyline (Correct)

D. Mirtazapine

Explanation: Tricyclic antidepressants (TCAs) such as amitriptyline potently block muscarinic acetylcholine receptors, leading to anticholinergic side effects including dry mouth, urinary retention, and constipation.

37. Which of the following first generation antipsychotic drug has high potency?

A. Chlorpromazine

B. Haloperidol (Correct)

C. Clozapine

D. Thioridazine

Explanation: Haloperidol is a high-potency first-generation (typical) antipsychotic characterized by strong dopamine D2 receptor antagonism and high potency (lower milligram dosage requirement), resulting in more extrapyramidal symptoms (EPS) and less sedation.

39. A 22-year-old man is admitted to the JUMC emergency room after an accidental overdose of a phenobarbital. He is unconscious and not considered a regular user of any medications or illicit drugs. Which treatment could be used to reverse the effect of the phenobarbital overdose?

A. Naloxone

B. Flumazenil

C. Diazepam

D. Ramelteon

E. None (Correct)

Explanation: There is no specific direct receptor antagonist reversal agent for barbiturates like phenobarbital (unlike benzodiazepines which are reversed by flumazenil or opioids by naloxone). Management involves supportive care, mechanical ventilation, and urine alkalinization with sodium bicarbonate to enhance excretion.

41. A 25-year-old man who has been drinking alcohol heavily is brought into the emergency department. At admission he is unconscious and his blood alcohol level is 275 mg/dL. What treatment should be provided to treat his intoxication?

A. Disulfiram

B. Naloxone

C. Flumazenil

D. Secobarbital

E. None (Supportive care) (Correct)

Explanation: Acute severe ethanol intoxication is managed primarily with supportive care (monitoring airway, breathing, circulation, and preventing aspiration). Specific antagonists like flumazenil or naloxone are not routinely recommended unless co-ingestions are suspected.